Solved selected problems of Modern Physics for Scientists and Engineers - John Taylor

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Chapter 5 - Quantization of Atomic Energy Levels

Solution. 5.1 Rydberg formula states that

$$\frac{1}{\lambda} = R\left(\frac{1}{n'^2} - \frac{1}{n^2}\right)$$

And with $n = 4$ and $n' = 3$ we have that

$$\frac{1}{\lambda} = 0.0110\left(\frac{1}{3^2} - \frac{1}{4^2}\right)$$
$$\frac{1}{\lambda} = 0.00053472$$
$$\lambda = 1870.1298\ nm$$

This wavelength correspond to infrared light. □

Solution. 5.3 The energy of photons is given by the equation

$$E_\gamma = hcR\left(\frac{1}{n'^2} - \frac{1}{n^2}\right) \quad \text{with} \quad n > n'$$

Then the upper limit of energies will be given for big values of n and n', hence if $n \to \infty$ and $n' \to \infty$ we get that

$$E_\gamma = hcR\left(0 - 0\right)$$
$$E_\gamma = 0\ eV$$

For the lower limit we need to consider big values of n' and small values of n hence if $n' \to \infty$ and $n = 1$ we get that

$$E_\gamma = hcR\left(0 - \frac{1}{1^2}\right)$$
$$E_\gamma = -1240\ [eV \cdot nm] \cdot 0.0110\ [nm^{-1}]$$
$$E_\gamma = -13.63\ eV$$

□

Solution. 5.8

(a) We know that $a_B = \hbar^2/ke^2m$ hence we can write Rydberg energy as

$$\begin{aligned} E_R &= \frac{ke^2}{2a_B} \\ &= ke^2\frac{ke^2m}{\hbar^2} \\ &= \frac{ke^2}{2}\frac{ke^2m}{\hbar^2} \\ &= \frac{m(ke^2)^2}{2\hbar^2} \end{aligned}$$

Which verifies that the two definition are equivalent.

(b) Given that $ke^2 = 1.44\ eV \cdot nm$ and $a_B = 0.0529\ nm$ we ge that

$$E_R = \frac{ke^2}{2a_B} = \frac{1.44\ eV \cdot nm}{2 \times 0.0529\ nm} = 13.61\ eV$$

For the other expression, multiplying top and bottom by c^2 and using that $mc^2 = 0.510\ MeV$ and $\hbar c = 197.3\ eV \cdot nm$ we get that

$$E_R = \frac{mc^2(ke^2)^2}{2(\hbar c)^2} = \frac{(0.510 \times 10^6\ eV) \times (1.44\ eV \cdot nm)^2}{2 \times (197.3\ eV \cdot nm)^2} = 13.58\ eV$$

□

Solution. 5.9 Let a charge q_1 with mass m in a circular orbit around a fixed charge q_2 such that q_2 has the opposite sign of q_1. Then the only force acting on q_1 is the Coulomb attraction of q_2 i.e.

$$F = \frac{kq_1q_2}{r^2}$$

Where k is the Coulomb force constant. Also, the centripetal acceleration of q_1 is $a = v^2/r$, hence by Newton's second law we have that

$$m\frac{v^2}{r} = \frac{kq_1q_2}{r^2}$$

Or

$$mv^2 = \frac{kq_1q_2}{r}$$

On the other hand, we know that the kinetic energy of q_1 is $K = \frac{1}{2}mv^2$ and the potential energy of q_1 in the field of q_2 is

$$U = -\frac{kq_1q_2}{r}$$

Since q_1 is of opposite sign to q_2. Thus, joining these equations we have that

$$K = -\frac{1}{2}U$$

And therefore the total energy is given by

$$E = K + U = -\frac{1}{2}U + U = \frac{1}{2}U$$

$\square$

Solution. 5.10

(a) From the angular momentum equation $mvr = n\hbar$ and Newton's second law $mv^2 = ke^2/r$ by replacing r we get that

$$mv\left(\frac{ke^2}{mv^2}\right) = n\hbar$$
$$\frac{ke^2}{v} = n\hbar$$
$$v = \frac{ke^2}{n\hbar}$$

Which is an expression for the electron's speed in the nth Bohr orbit.

(b) Given that the speed is an inverse function of n then the highest speed is going to happen when $n = 1$ i.e.

$$v_1 = \frac{ke^2}{\hbar}$$

And hence for the electron we have that

$$\begin{aligned} v_1 &= \frac{8.99 \times 10^9 \ Nm^2/C^2 \cdot (1.602 \times 10^{-19} \ C)^2}{1.054 \times 10^{-34} \ Js} \\ &= 2,188,991.64 \ m/s \end{aligned}$$

(c) Assuming that $c = 299,792,458 \ m/s$ then the ratio

$$\alpha = \frac{v_1}{c} = \frac{ke^2}{\hbar^2}$$

is given by

$$\alpha = \frac{v_1}{c} = \frac{2,188,991.64 \ m/s}{299,792,458 \ m/s} = 0.0073016$$

And therefore

$$\alpha = 0.0073016 \approx \frac{1}{137} = 0.0072992$$

$\square$

Solution. 5.13

(a) We know that E_R is proportional to m_e so for the muonic hydrogen atom we replace m_e by $m_\mu = 207 \cdot m_e$ hence $E_R = 207 \cdot 13.6\ eV = 2815.2\ eV$ so the energy of the first Bohr orbit is

$$E_1 = -2815.2\ eV$$

The radius of the orbit is inversely proportional to the mass m_e so replacing m_e with $m_\mu = 207 \cdot m_e$ we get that

$$r = \frac{n^2 a_B}{207}$$

Hence the radius of the first Bohr orbit is

$$r = \frac{0.0529}{207} = 0.000255\ nm$$

(b) The energy of the Lymann α line is

$$\begin{aligned} E_\alpha &= E_2 - E_1 \\ &= -\frac{E_R}{2^2} + \frac{E_R}{1^2} \\ &= -703.8\ eV + 2815.2\ eV \\ &= 2111.39\ eV \end{aligned}$$

Hence the wavelength is

$$\lambda_\alpha = \frac{hc}{E_\alpha} = \frac{1240\ eV \cdot nm}{2111.4\ eV} = 0.587\ nm$$

This electromagnetic wavelength corresponds to X-rays.

□

Solution. 5.14

(a) We know the atomic diameter is given by $d = 2n^2 a_B$ so if we set $d = D = 3\ nm$ at normal densities the largest n we can find is

$$n = \sqrt{\frac{D}{2a_B}} = \sqrt{\frac{3\ nm}{2 \cdot 0.0529\ nm}} = 5.32 \approx 5$$

(b) Given that D is proportional to $P^{-1/3}$ then we can write that

$$D = kP^{-1/3}$$

For some constant k and we know that $D = 3\ nm$ at a pressure of $P = 1\ atm$ then must be that

$$k = 3\ nm \cdot \sqrt[3]{atm}$$

Also, we know that $n = \sqrt{\frac{D}{2a_B}}$ so for a pressure of $1/1000\ atm$ the largest n we can have is

$$n = \sqrt{\frac{kP^{-1/3}}{2a_B}} = \sqrt{\frac{3 \cdot 10}{2 \cdot 0.0529\ nm}} = 16.83 \approx 17$$

(b) First, we determine the equation of the pressure P in terms of n then

$$\sqrt[3]{P} = \frac{k}{2n^2 a_B}$$
$$P = \left(\frac{k}{2n^2 a_B}\right)^3$$

If we set $n = 100$ then the pressure in these experiments is

$$P = \left(\frac{3}{2 \cdot 10000 \cdot 0.0529}\right)^3 = 2.279 \times 10^{-8}\ atm$$

$\square$

Solution. 5.15 The energy of a photon in the Lymann α line of Fe^{25+} is

$$\begin{aligned} E_\alpha &= Z^2 E_R \left(\frac{1}{n'^2} - \frac{1}{n^2} \right) \\ &= 26^2 \cdot 13.6 \cdot \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \\ &= 6895.2 \ eV \end{aligned}$$

And hence the wavelength associated to this energy is

$$\lambda_\alpha = \frac{hc}{E_\alpha} = \frac{1240}{6895.2} = 0.1798 \ nm$$

This wavelength corresponds to X-ray radiation. □

Solution. 5.16 The radius of the $n = 1$ orbit in the O^{7+} ion is given by

$$r = n^2 \frac{a_B}{Z} = \frac{0.0529}{8} = 0.00661 \ nm$$

The energy of a photon in the Lymann α line of O^{7+} is

$$\begin{aligned} E_\alpha &= Z^2 E_R \left(\frac{1}{n'^2} - \frac{1}{n^2} \right) \\ &= 8^2 \cdot 13.6 \cdot \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \\ &= 652.8 \ eV \end{aligned}$$

And hence the wavelength associated to this energy is

$$\lambda_\alpha = \frac{hc}{E_\alpha} = \frac{1240}{652.8} = 1.899 \ nm$$

□

Solution. 5.17

(a) Considering the proton as fixed then the energy of a hydrogen atom is

$$E_R = \frac{m(ke^2)^2}{2\hbar^2} = 13.60569\ eV$$

but if we take into account the movement of the nucleus via the reduced mass we get that

$$E_R' = \frac{E_R}{1 + m/m_{nuc}} = \frac{13.60569}{1 + 9.10938 \times 10^{-31}/1.67262 \times 10^{-27}} = \\ = 13.59828\ eV$$

Then the percent error in the energy levels is

$$\left| \frac{E_R' - E_R}{E_R} \right| \times 100 = 0.05446\%$$

(b) For a muonic hydrogen, if the proton is fixed we get that

$$E_R^\mu = \frac{m_\mu}{m} E_R = \frac{1.88353 \times 10^{-28}}{9.10938 \times 10^{-31}} \cdot 13.60569 = 2813.22387\ eV$$

But if the proton is moving we have that

$$E_R^{\mu\prime} = \frac{E_R^\mu}{1 + m_\mu/m_{nuc}} = \frac{2813.22387}{1 + 1.88353 \times 10^{-28}/1.67262 \times 10^{-27}} = \\ = 2528.49155\ eV$$

Finally, the percent error in the energy levels for the muonic hydrogen is

$$\left| \frac{E_R^{\mu\prime} - E_R^\mu}{E_R^\mu} \right| \times 100 = 10.12121\%$$

□

Solution. 5.18 We want to find the ground-state energy of positronium, this energy will differ from the ground-state energy of the hydrogen only in the mass used to compute it so we can write that

$$E_p = \frac{\mu}{m_e} E_e$$

Where $\mu = m_e/(1 + m_e/m_{nuc})$ is the reduced mass for the positronium. Then

$$\begin{aligned}\mu &= \frac{m_e}{1 + m_e/m_{nuc}} \\ &= \frac{9.1093 \times 10^{-31}}{1 + (9.1093 \times 10^{-31}/9.1093 \times 10^{-31})} \\ &= 4.5546 \times 10^{-31}\ kg\end{aligned}$$

Hence

$$E_p = \frac{4.5546 \times 10^{-31}}{9.1093 \times 10^{-31}} \times -13.7\ eV = \frac{-13.7\ eV}{2} = -6.8\ eV$$

□

Solution. 5.20

(a) We know that the orbital radius is given by

$$r = n^2 \frac{a_B}{Z}$$

Where a_B was calculated using the electron mass m_e so we can compute what the orbital radius for a π^- captured in the $n = 1$ orbit by a carbon nucleus as follows

$$r = n^2 \frac{a_B}{Z} \frac{m_e}{m_\pi} = n^2 \frac{a_B}{Z} \frac{m_e}{273 m_e} = n^2 \frac{a_B}{273 Z}$$

Hence

$$r = \frac{0.0529}{273 \times 6} = 3.2295 \times 10^{-5}\ nm$$

(a) Given that the carbon nucleus has a radius $R \approx 3 \times 10^{-6}\ nm$ the π^- orbit can be formed.

(c) For a lead nucleus we have that the π^- orbital radius is

$$r = \frac{0.0529}{273 \times 82} = 2.3630 \times 10^{-6}\ nm$$

But the lead nucleus has a radius of $R \approx 7 \times 10^{-6}\ nm$ so the π^- would have a smaller orbit and hence the orbit cannot be formed.

□

Solution. 5.21

(a) Assuming the center of mass is at the origin of the system by the equation for the coordinates of the center of mass we have that $m_e r_e - m_p r_p = 0$ or $m_e r_e = m_p r_p$ also we know that $r = r_e + r_p$ hence

$$\begin{aligned} r &= r_e + \frac{m_e}{m_p} r_e \\ r &= r_e\left(1 + \frac{m_e}{m_p}\right) \\ r_e &= \frac{m_p}{m_p + m_e} r \end{aligned}$$

And for r_p we get that

$$\begin{aligned} r &= \frac{m_p}{m_e} r_p + r_p \\ r &= r_p\left(\frac{m_p}{m_e} + 1\right) \\ r_p &= \frac{m_e}{m_p + m_e} r \end{aligned}$$

(b) The total kinetic energy is given by

$$\begin{aligned} K &= K_e + K_p \\ &= \frac{1}{2} m_e (r_e \omega)^2 + \frac{1}{2} m_p (r_p \omega)^2 \\ &= \frac{1}{2} \frac{m_e m_p^2}{(m_p + m_e)^2} r^2 \omega^2 + \frac{1}{2} \frac{m_p m_e^2}{(m_p + m_e)^2} r^2 \omega^2 \\ &= \frac{1}{2} \frac{m_e m_p (m_p + m_e)}{(m_p + m_e)^2} r^2 \omega^2 \\ &= \frac{1}{2} \frac{m_e m_p}{m_p + m_e} r^2 \omega^2 \end{aligned}$$

Hence $K = \frac{1}{2} \mu r^2 \omega^2 a$ where μ is the reduced mass.

(c) Applying Newton's law to the electron in addition to Coulomb's law we get that

$$\frac{ke^2}{r^2} = m_e \frac{v_e^2}{r_e}$$

Replacing the formulas we have for r_e and v_e we get that

$$\begin{aligned} \frac{ke^2}{r^2} &= m_e r_e \omega^2 \\ \frac{ke^2}{r^2} &= m_e \frac{m_p}{m_p + m_e} r \omega^2 \\ \frac{ke^2}{r^2} &= \mu \omega^2 r \end{aligned}$$

(d) We know that the potential energy of an electron in the field of a proton is $U = -ke^2/r$ hence from the formula derived in part (c) we have that

$$U = -\frac{ke^2}{r} = -\mu\omega^2 r^2$$

Also, we know that $K = 1/2\mu r^2\omega^2$ so $K = -U/2$ and therefore the total energy is given by

$$E = K + U = \frac{U}{2} = -\frac{1}{2}\frac{ke^2}{r}$$

(e) The total angular momentum is

$$L = L_e + L_p$$

Hence

$$\begin{aligned} L &= m_e v_e r_e + m_p v_p r_p \\ &= m_e r_e^2 \omega + m_p r_p^2 \omega \\ &= \frac{m_e m_p^2}{(m_p + m_e)^2} r^2 \omega + \frac{m_p m_e^2}{(m_p + m_e)^2} r^2 \omega \\ &= \frac{m_e m_p (m_p + m_e)}{(m_p + m_e)^2} r^2 \omega \\ &= \mu r^2 \omega \end{aligned}$$

(f) Let $L = n\hbar$ then by the formula derived in (e) we have that

$$\begin{aligned} \mu r^2 \omega &= n\hbar \\ \omega &= \frac{n\hbar}{\mu r^2} \end{aligned}$$

Also, we know that $\mu r \omega^2 = ke^2/r^2$ so replacing ω we get that

$$\begin{aligned} \mu r \left(\frac{n\hbar}{\mu r^2}\right)^2 &= \frac{ke^2}{r^2} \\ \frac{(n\hbar)^2}{\mu r^3} &= \frac{ke^2}{r^2} \\ \frac{(n\hbar)^2}{\mu r} &= ke^2 \\ r &= \frac{n^2\hbar^2}{\mu k e^2} \end{aligned}$$

Which gives us the allowed radii.

Now replacing the value of r in the equation we have for E we get that

$$\begin{aligned} E &= -\frac{1}{2}\frac{ke^2}{\frac{n^2\hbar^2}{\mu ke^2}} \\ &= -\frac{1}{2}\frac{\mu(ke^2)^2}{n^2\hbar^2} \end{aligned}$$

So defining

$$E_R = \frac{\mu(ke^2)^2}{2\hbar^2}$$

We get that $E = -E_R/n^2$.

(g) The reduced mass is

$$\mu = \frac{938.27208 \times 0.51099}{938.27208 + 0.51099} = 0.51072\ MeV/c^2$$

Then the energy of the ground state of hydrogen using the reduced mass give us

$$\begin{aligned} E_R^{\mu} &= \frac{0.51072 \times 10^6 \cdot (1.44)^2}{2 \cdot (6.582 \times 10^{-16})^2} \cdot \frac{1}{(2.998 \times 10^{17})^2} \\ &= 13.59874\ eV \end{aligned}$$

But the energy of the ground state for the fixed-proton is

$$\begin{aligned} E_R^{\mu} &= \frac{0.51099 \times 10^6 \cdot (1.44)^2}{2 \cdot (6.582 \times 10^{-16})^2} \cdot \frac{1}{(2.998 \times 10^{17})^2} \\ &= 13.60593\ eV \end{aligned}$$

□

Solution. 5.22 From equation (5.37) we know that

$$E_\gamma = \frac{3}{4}(Z-\delta)^2 E_R$$

Hence

$$hf = \frac{3}{4}(Z-\delta)^2 E_R$$
$$\sqrt{f} = \sqrt{\frac{3E_R}{4h}}(Z-\delta)$$

Then the slope of the graph of $\sqrt{f}$ against Z is

$$\sqrt{\frac{3E_R}{4h}} = \sqrt{\frac{3 \times 13.6}{4 \times 4.135 \times 10^{-15}}} = 49,666,359.63\ \sqrt{Hz}$$

And from the graph we see that the predicted slope is

$$\frac{\Delta\sqrt{f}}{\Delta Z} = \frac{(20 \times 10^8 - 10 \times 10^8)}{40-20} = \frac{10 \times 10^8}{20} = 50,000,000\ \sqrt{Hz}$$

$\square$

Solution. 5.25 The radius of the $n = 1$ orbit of the innermost electron in the lead atom is given by

$$\begin{aligned} r &= n^2 \frac{a_B}{Z} \\ &= 1^2 \frac{0.0529 \times 10^{-9}}{82}\ m \\ &= 6.451 \times 10^{-13}\ m \end{aligned}$$

Compared to the lead nucleus radius is quite far from the nucleus. □

Solution. 5.26

(a) Considering that $m_\mu = 207\ m_e$ then the radius of the muon's orbit differs from an electron orbit by a factor of 207 then

$$r = n^2 \frac{a_B}{207\ Z} = 1^2 \frac{0.0529}{207 \cdot 47} = 5.437\ fm$$

The muon's orbital radius is very close to the nuclear radius so, yes, it is a good approximation to ignore the atomic electrons when considering the muon.

(b) When a muon drops from $n = 2$ to $n = 1$ the energy emitted is given by

$$E_\gamma = Z^2 \cdot 207 \cdot E_R \left(1 - \frac{1}{4}\right)$$

Where E_R depends on m_e so for a muon we multiply it by 207. Then

$$E_\gamma = 47^2 \cdot 207 \cdot 13.6 \left(1 - \frac{1}{4}\right) = 4664082.6\ eV$$

The wavelength associated to this energy is

$$\lambda = \frac{hc}{E_\gamma} = \frac{1240}{4664082.6} = 2.65 \times 10^{-13}\ m$$

Which is a gamma ray wavelength.

□

Solution. 5.27

(a) From conservation of momentum we have that

$$\begin{aligned} v_0 m &= v_r M - v_f m \\ v_r &= \frac{m}{M}(v_0 + v_f) \end{aligned}$$

Where v_r is the recoil velocity and v_f is the electron's final velocity. Then the recoil kinetic energy is

$$\begin{aligned} K_r &= \frac{1}{2} M v_r^2 \\ &= \frac{1}{2}\frac{m^2}{M}(v_0 + v_f)^2 \\ &= \frac{1}{2}\frac{m^2}{M}(v_0^2 + 2v_0 v_f + v_f^2) \end{aligned}$$

But if we suppose $v_0 \approx v_f$ we get that

$$K_r \approx \frac{1}{2}\frac{m^2}{M} 4v_0^2 = 4\frac{m}{M}K_0$$

(b) The maximum possible loss of kinetic energy for a 3eV electron is the same as the maximum possible recoiling energy the atom can have then by applying the approximation from part (a) we get that

$$K = 4 \cdot \frac{9.10938 \times 10^{-31}}{3.33091 \times 10^{-25}} \cdot 3 = 3.281 \times 10^{-5} eV = 32.81 \mu eV$$

□